

Video Transcript - Part 5

Slide 1

Hey everyone, welcome back. For part four, we're comparing two AWS streaming services that often get mixed up, Amazon Kinesis Data Streams and Amazon MSK. Both handle real time data streaming, and both can feed straight into a machine learning pipeline, but they take pretty different approaches under the hood, and picking the right one matters for how your pipeline scales, integrates, and evolves down the line.

Slide 2

Here's the plan. We'll start with what these two services actually have in common. Then we'll look at what Kinesis Data Streams is, followed by what Amazon MSK is, walk through the key differences between them, and finish with clear guidance on when to reach for each one.

Slide 3

Before we get into the differences, it's worth noting what these two services actually have in common. Both are fully managed, real time streaming services on AWS. Both durably store streaming data for a retention window, so consumers can process it independently and even replay it if needed. And both can feed into the exact same downstream services, things like Lambda, S3, Redshift, and machine learning pipelines. So the real decision here isn't about capability, it comes down to the technology underneath, and how much control you want over it.

Slide 4

Let's start with Amazon Kinesis Data Streams. It's a fully managed, real time streaming service built natively for AWS. Data flows through shards, and each shard gives you a fixed unit of read and write capacity. You can run it in provisioned mode, where you set the shard count yourself, or on-demand mode, where AWS scales shards for you automatically as your traffic changes. By default, data is retained for twenty four hours, but you can extend that all the way up to a full year. To consume the data, you'd typically use the Kinesis SDK, the Kinesis Client Library, AWS Lambda, or a service like Kinesis Data Analytics.

Slide 5

Now, Amazon MSK, which stands for Managed Streaming for Apache Kafka. This is a fully managed version of Apache Kafka, the popular open source streaming platform, so you get real Kafka running under the hood. Because it speaks the standard Kafka protocol, existing Kafka producers and consumers just work without changes. You can run it as Provisioned, where you size the brokers and storage yourself, or as MSK Serverless, which scales automatically. And since it's genuine Kafka, it's compatible with the whole Kafka ecosystem, tools like Kafka Connect, Kafka Streams, and ksqlDB. It's a natural fit if your team already knows Kafka, or you're migrating an existing Kafka cluster into AWS.

Slide 6

So what are the real differences here? Kinesis runs on AWS's own proprietary technology, while MSK runs genuine, open source Apache Kafka. Kinesis uses its own API and SDK, while MSK speaks the standard Kafka protocol. Kinesis scales through shards, either manually or on-demand, while MSK scales through brokers and partitions, or through MSK Serverless. Kinesis integrates tightly with other AWS services out of the box, while MSK plugs into the broader Kafka ecosystem. Operationally, Kinesis abstracts away almost everything, while MSK gives you more control, at the cost of more configuration. And one more thing worth noting, Kafka is portable across cloud providers, so MSK workloads aren't locked into AWS the way Kinesis workloads are.

Slide 7

So when should you reach for Kinesis Data Streams? If you're building an AWS native application and want the simplest path to streaming, Kinesis is a great fit. It's also the right call if you want tight, out of the box integration with services like Lambda, Firehose, or Kinesis Data Analytics. And if your team doesn't have existing Kafka expertise, Kinesis lets you skip that learning curve entirely, since AWS handles most of the operational overhead for you.

Slide 8

On the other hand, reach for Amazon MSK when you're already running Apache Kafka somewhere and want to move to AWS without rewriting your producers and consumers. It's also the better choice if you need specific Kafka ecosystem tools, like Kafka Connect for integrations or ksqlDB for stream processing. And if portability matters to you, avoiding being locked into a single cloud provider, MSK keeps that door open, since it's still standard Kafka underneath. If you also need fine grained control over brokers, partitions, and configuration, MSK gives you that flexibility too.

Slide 9

So to recap, Kinesis Data Streams is AWS's own fully managed streaming service, simple to use and tightly integrated with the AWS ecosystem. Amazon MSK is a fully managed version of Apache Kafka, giving you the power and portability of open source Kafka with less operational burden. Choose Kinesis when you want simplicity and native AWS integration, and choose MSK when you need Kafka compatibility, its ecosystem, or portability across clouds. Thanks for watching, and I'll see you in the next one.